MA1522 Linear Algebra for Computing

Lesson 8: Linear Spans and Subspaces

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson8` (e.g. `A1234567Z_lesson8.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 8**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 3.3 Comparing Two Spans

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $T = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$ be sets of vectors in $\mathbb{R}^n$.

Theorem

$\forall\, \mathbf{v}_i \in T, \quad \mathbf{v}_i \in \mathrm{Span}(S)$.

$$\mathrm{span}(T) \subseteq \mathrm{span}(S) \iff \left(\begin{array}{ccc|ccc} \mathbf{u}_1 & \cdots & \mathbf{u}_k & \mathbf{v}_1 & \cdots & \mathbf{v}_m \end{array} \right) \text{ is consistent.}$$

So to check whether $\mathrm{span}(S) = \mathrm{span}(T)$, reduce twice:

$$\left(\begin{array}{c|c} S & T \end{array} \right) \text{ consistent} \iff \mathrm{span}(T) \subseteq \mathrm{span}(S),$$

$$\left(\begin{array}{c|c} T & S \end{array} \right) \text{ consistent} \iff \mathrm{span}(S) \subseteq \mathrm{span}(T).$$

Question 1

Alice and Bob each solve the same homogeneous system and write the general solution as

Alice: $s\begin{pmatrix}1\\1\\0\end{pmatrix} + t\begin{pmatrix}-1\\0\\1\end{pmatrix}, \quad s,t \in \mathbb{R},$ $= \operatorname{span}\left(\left\{\begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}\right)$

Bob: $p\begin{pmatrix}1\\2\\1\end{pmatrix} + q\begin{pmatrix}1\\3\\2\end{pmatrix}, \quad p,q \in \mathbb{R}.$ $= \operatorname{span}\left(\left\{\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}1\\3\\2\end{pmatrix}\right\}\right)$

Do Alice and Bob describe the same set of solutions?

Question 1 — solution

$\text{span}\left(\left\{\begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}\right)$ $\quad$ $\text{span}\left(\left\{\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}1\\3\\2\end{pmatrix}\right\}\right)$

S $\quad$ T

$$\left(\begin{array}{cc|cc} 1 & -1 & 1 & 1 \\ 1 & 0 & 2 & 3 \\ 0 & 1 & 1 & 2 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{cc|cc} 1 & 0 & 2 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

consistent $\Rightarrow \text{span}(S) \supseteq T$.

$\text{span}(S) \supseteq \text{span}(T)$.

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & -1 \\ 2 & 3 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

consistent $\Rightarrow \text{span}(T) \supseteq \text{span}(S)$.

We conclude that $\text{span}(S) = \text{span}(T)$,

so the answers of Alice and Bob agree.

Section 3.4 Subspaces

We say $V \subseteq \mathbb{R}^n$ is a subspace if
- it is a subset of $\mathbb{R}^n$
- itself is a vector space (under the same addition, scalar mult).

Definition
A subset V of $\mathbb{R}^n$ is a **subspace** if

(i) $\mathbf{0} \in V$;

(ii) V is closed under scalar multiplication: $\alpha\mathbf{v} \in V$ for all $\mathbf{v} \in V$ and $\alpha \in \mathbb{R}$;

(iii) V is closed under addition: $\mathbf{u} + \mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$.

Equivalently, V is a subspace if

(i′) V is nonempty;

(ii′) V is closed under linear combinations: $\alpha\mathbf{u} + \beta\mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$ and all $\alpha, \beta \in \mathbb{R}$.

To show that V is not a subspace, exhibit one of the following: $\mathbf{0} \notin V$; a vector $\mathbf{v} \in V$ and a scalar α with $\alpha\mathbf{v} \notin V$; or vectors $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{u} + \mathbf{v} \notin V$.

Sections 3.3 & 3.4 Properties of Linear Spans

Theorem

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a set of vectors in $\mathbb{R}^n$.

(i) The zero vector $\mathbf{0}$ is in span(S).

(ii) span(S) is closed under scalar multiplication: if $\mathbf{u} \in \text{span}(S)$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{u} \in \text{span}(S)$.

(iii) span(S) is closed under addition: if $\mathbf{u}, \mathbf{v} \in \text{span}(S)$, then $\mathbf{u} + \mathbf{v} \in \text{span}(S)$.

Namely, span(S) is a subspace.

An equivalent definition of Span(S) is that it is the smallest subspace of $\mathbb{R}^n$ that contains S.

Theorem (Section 3.4: equivalent definition for subspaces)

A subset $V \subseteq \mathbb{R}^n$ is a subspace if and only if it is a linear span, $V = \text{span}(S)$, for some finite set $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ of vectors in $\mathbb{R}^n$.

Question 2

Determine whether each of the following subsets of $\mathbb{R}^3$ is a subspace.

1. $V_1 = \{ (x, y, z)^T \in \mathbb{R}^3 \mid x = 2y \}$ Yes
2. $V_2 = \{ (x, y, z)^T \in \mathbb{R}^3 \mid x \geq 0 \}$ No. because $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \in V_2$ but $(-1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$ (≥ 0, $\neq 0$) $\notin V_2$, so (ii) fails.
3. $V_3 = \{ (x, y, z)^T \in \mathbb{R}^3 \mid xy = 0 \}$ No

If it is a subspace, express it as a linear span. If it is not, say which of the three conditions fails, and give an explicit vector or pair of vectors that shows it.

Question 2 — solution: V_1 and V_2

$V_1 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 \;\middle|\; \boxed{x = 2y} \right\}$: general solution of $x - 2y = 0$.

Step 1. check V_1 is a subspace

- (i') : V_1 is nonempty : $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \in V_1$
- (ii') : $\forall \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \in V_1$, $\alpha, \beta \in \mathbb{R}$, check $\alpha \underbrace{\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}}_{v} + \beta \underbrace{\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}}_{w} \overset{?}{\in} V_1$

$$\alpha \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \beta \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} \alpha x_1 + \beta x_2 \\ \alpha y_1 + \beta y_2 \\ \alpha z_1 + \beta z_2 \end{pmatrix}$$

check $\alpha x_1 + \beta x_2 \overset{?}{=} 2(\alpha y_1 + \beta y_2)$

yes, because $\begin{cases} x_1 = 2y_1 \\ x_2 = 2y_2 \end{cases}$, since $v, w \in V_1$

Step 2. Write V_1 as span.

To write V_1 as a span, we solve

$\{ x - 2y = 0 \quad (1 \;\; -2 \;\; 0 \;|\; 0)$ general sol. $\begin{cases} x = 2s \\ y = s \\ z = t \end{cases}$ $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = s \cdot \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ s.t. $\in \mathbb{R}$.

non-pivot.

So $V_1 = \operatorname{span}\left(\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\} \right)$

Question 2 — solution: V_3

In fact. $V_2 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \;\middle|\; x \geq 0 \right\}$ is closed under addition but not closed under scalar mult.

- If $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} \in V_2$. $\begin{pmatrix} x_1 \geq 0 \\ x_2 \geq 0 \end{pmatrix}$.

$\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix} = \begin{pmatrix} x_1+x_2 \\ y_1+y_2 \\ z_1+z_2 \end{pmatrix}$. $x_1+x_2 \geq 0$. $\in V_2$.

- not closed under $(-1)\cdot V$.

$V_3 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \;\middle|\; xy = 0 \right\}$.

not linear

usually not preserving addition.

not closed under addition:

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

$\in V_3$ $\in V_3$ $\notin V_3$.

- One can check that it is closed under scalar mult.

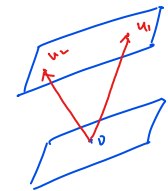
Section 3.4 Affine Spaces

Let $A\mathbf{x} = \mathbf{b}$ be consistent, let $\mathbf{u}$ be a **particular solution**, $A\mathbf{u} = \mathbf{b}$, and let $V = \{\, \mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \mathbf{0} \,\}$ be the solution space of the associated homogeneous system.

Theorem (Affine Space)

The solution set of $A\mathbf{x} = \mathbf{b}$ is

$$W = \mathbf{u} + V := \{\, \mathbf{u} + \mathbf{v} \mid \mathbf{v} \in V \,\}.$$



V : sol. of the homogeneous part.
a subspace.

In general an affine space $\mathbf{u} + V$ is not a subspace: it is one exactly when $\mathbf{u} \in V$. Here $\mathbf{u} \in V$ means $A\mathbf{u} = \mathbf{0}$, that is $\mathbf{b} = \mathbf{0}$, and then $W = V$. For $\mathbf{b} \neq \mathbf{0}$ we get $\mathbf{0} \notin W$, and W is V shifted away from the origin by $\mathbf{u}$: the two are parallel.

Example

Let $\mathbf{u}_1 + W_1$ and $\mathbf{u}_2 + W_2$ be two affine spaces inside $\mathbb{R}^n$. Here $\mathbf{u}_1, \mathbf{u}_2$ are two vectors of $\mathbb{R}^n$ and W_1, W_2 are subspaces of $\mathbb{R}^n$. True or false?

1. If $\mathbf{u}_1 + W_1 = \mathbf{u}_2 + W_2$ as subsets of $\mathbb{R}^n$, then $\mathbf{u}_1 = \mathbf{u}_2$. False.
2. If $\mathbf{u}_1 + W_1 = \mathbf{u}_2 + W_2$ as subsets of $\mathbb{R}^n$, then $W_1 = W_2$. True

1. A counter example $\begin{pmatrix}1\\0\\0\end{pmatrix} + \operatorname{span}\left(\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}\right\}\right) = \begin{pmatrix}0\\0\\0\end{pmatrix} + \operatorname{span}\left(\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}\right\}\right)$

$\mathbf{u}_1 \notin W_1$. $\quad \mathbf{u}_2 \in W_1$.

2. Step 1. $u_2 - u_1 \in W_1$.

because $u_2 + 0 \in u_2 + W_2 = u_1 + W_1$ $\Rightarrow u_2 - u_1 = w_1 \in W_1$.

$u_1 + w_1$ for some $w_1 \in W_1$.

Step 2. $W_2 \subseteq W_1$: $\forall\, w_2 \in W_2$, $u_2 + w_2 \in u_2 + W_2 = u_1 + W_1$ $\Rightarrow w_2 = (u_1 - u_2) + w_1 \Rightarrow w_2 \in W_1$.

$u_1 + w_1$ for some $w_1 \in W_1$. $\quad u_1 - u_2 \in W_1$, $w_1 \in W_1$.

Challenge

Let W be a subspace of $\mathbb{R}^n$ and let $\mathbf{u}_1, \mathbf{u}_2 \in \mathbb{R}^n$. Show that

$$\mathbf{u}_1 + W = \mathbf{u}_2 + W \iff \mathbf{u}_1 - \mathbf{u}_2 \in W.$$

Question 3

Alice and Bob each solve the same system $A\mathbf{x} = \mathbf{b}$, with $\mathbf{b} \neq \mathbf{0}$, and write the general solution as

Alice: $\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$,

Bob: $\begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + p \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + q \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$,

with $s, t, p, q \in \mathbb{R}$.

The same as in Question 1.

Do Alice and Bob describe the same set of solutions?

Question 3 — solution

$$\underbrace{\begin{pmatrix}3\\0\\0\end{pmatrix}}_{u_1} + \operatorname{span}\left(\left\{\underbrace{\begin{pmatrix}1\\1\\0\end{pmatrix}}_{w_1=w},\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}\right) \overset{?}{=} \underbrace{\begin{pmatrix}0\\-2\\0\end{pmatrix}}_{u_2} + \operatorname{span}\left(\left\{\underbrace{\begin{pmatrix}1\\2\\1\end{pmatrix}}_{w_2=w},\begin{pmatrix}1\\3\\2\end{pmatrix}\right\}\right).$$

Step 1 Example 2: If $u_1+W_1 = u_2+W_2$, then $W_1 = W_2$.

Equivalently, if $W_1 \neq W_2$, then $u_1+W_1 \neq u_2+W_2$.

(aim to check this first.)

In fact, we have checked that $W_1 = W_2$ in Question 1. ✓

Step 2 Now we use the challenge. $u_1+W = u_2+W \iff u_1-u_2 \in W$.

Then $u_1+W_1 = u_2+W_2 \iff u_1-u_2 \in W$

$u_1-u_2 = \begin{pmatrix}3\\2\\0\end{pmatrix}$, $W = \operatorname{span}\left(\left\{\begin{pmatrix}1\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}\right)$.

$$\left(\begin{array}{cc|c}1&-1&3\\1&0&2\\0&1&0\end{array}\right) \xrightarrow{\text{ref}} \left(\begin{array}{cc|c}1&0&0\\0&1&0\\0&0&1\end{array}\right)$$

not consistent

I.e. $u_1-u_2 \notin W_1$

Next lesson

Conclusion: Two solution sets do not agree.

- **Sections 3.4 & 3.5 Subspaces and Linear Independence**

Please watch the video clips for these sections beforehand.